Functional Requirements

Module 1: Calendar & Email Parsing

FR1.1

The system shall allow users to securely sign in using their Google account.

FR1.2

The system shall synchronize events from the user's Google Calendar automatically. **FR1.3**

The system shall retrieve emails related to meetings, assignments, deadlines, and appointments.

FR1.4

The system shall extract important event information (title, date, time, location, deadline, participants) from emails using Natural Language Processing (NLP).

FR1.5

The system shall automatically create calendar events from extracted email information.

FR1.6

The system shall detect and prevent duplicate events during synchronization.

FR1.7

The system shall classify extracted events into categories such as Study, Work, Personal, Health, and Meeting.

FR1.8

The system shall allow users to edit or delete automatically extracted events before saving them.

FR1.9

The system shall notify users whenever a new important event is detected.

FR1.10

The system shall continuously synchronize calendar and email updates in the background.

FR1.11

The system shall assign an initial priority hint for extracted events.

Module 2: Scheduler

FR2.1

The system shall allow users to create, edit, and delete tasks.

FR2.2

The system shall allow users to specify task priority, deadline, estimated duration, and category.

FR2.3

The system shall automatically generate an optimized daily schedule based on task priority, deadlines, and available time.

FR2.4

The system shall automatically generate an optimized weekly schedule.

FR2.5

The system shall prevent overlapping tasks within the generated schedule.

FR2.6

The system shall consider fixed events (meetings, classes, exams) before scheduling flexible tasks.

FR2.7

The system shall recommend the best time slot for each task.

FR2.8

The system shall allow users to manually rearrange scheduled tasks.

FR2.9

The system shall save all schedule modifications.

FR2.10

The system shall display the generated schedule in daily, weekly, and monthly calendar views.

FR2.11

The system shall classify each task as Fixed or Flexible based on its category.

FR2.12

The system shall avoid moving Fixed tasks during optimization.

Module 3: Energy Score

FR3.1

The system shall calculate the user's daily Energy Score based on sleep duration, workload, and daily activities.

FR3.2

The system shall update the Energy Score whenever new health or activity information becomes available.

FR3.3

The system shall classify the user's energy level as High, Medium, or Low.

FR3.4

The system shall recommend suitable task intensity according to the user's current Energy Score.

FR3.5

The system shall notify users when their Energy Score becomes critically low.

FR3.6

The system shall display the current Energy Score on the dashboard.

FR3.7

The system shall store historical Energy Score records.

FR3.8

The system shall generate weekly Energy Score reports.

FR3.9

The system shall use the Energy Score as an input for scheduling decisions.

FR3.10

The system shall update scheduling recommendations whenever the Energy Score changes.

FR3.11

The system shall estimate accumulated fatigue throughout the day.

FR3.12

The system shall consider recovery periods before calculating Energy Score.

FR3.13

The system shall estimate the user's peak productive hours.

Module 4: Attention

FR4.1

The system shall calculate the user's Attention Score during the working day.

FR4.2

The system shall estimate attention level using user activity and interaction patterns.

FR4.3

The system shall classify attention level into High, Medium, or Low.

FR4.4

The system shall recommend focus sessions for high-priority tasks.

FR4.5

The system shall recommend break periods when attention decreases.

FR4.6

The system shall notify users when prolonged low attention is detected.

FR4.7

The system shall display daily and weekly Attention Score trends.

FR4.8

The system shall store historical Attention Score records.

FR4.9

The system shall use the Attention Score in scheduling decisions.

FR4.10

The system shall update recommendations whenever the Attention Score changes.

FR4.11

The system shall use Screen Time metrics when calculating Attention Score.

FR4.12

The system shall shorten focus sessions when attention decreases significantly.

Module 5: Decision Engine

FR5.1

The system shall evaluate task priority before generating recommendations.

FR5.2

The system shall consider Energy Score when selecting the next task.

FR5.3

The system shall consider Attention Score when selecting the next task.

FR5.4

The system shall recommend the most suitable task based on multiple decision criteria. **FR5.5**

The system shall recommend postponing low-priority tasks when required.

FR5.6

The system shall recommend suitable break times based on workload and attention level. **FR5.7**

The system shall generate personalized productivity recommendations.

FR5.8

The system shall explain the reason behind every recommendation.

FR5.9

The system shall continuously evaluate scheduling decisions after every schedule update.

FR5.10

The system shall improve future recommendations based on user behavior and previous scheduling history.

FR5.11

The system shall consider task deadlines during recommendation generation.

FR5.12

The system shall normalize decision criteria before calculating the Decision Score.

Module 6: Dynamic Rescheduling

FR6.1

The system shall automatically detect schedule conflicts.

FR6.2

The system shall automatically reschedule affected tasks when new events are added.

FR6.3

The system shall automatically reschedule tasks when deadlines change.

FR6.4

The system shall preserve completed tasks during rescheduling.

FR6.5

The system shall preserve task priorities whenever possible.

FR6.6

The system shall notify users before applying major schedule changes.

FR6.7

The system shall allow users to accept or reject the proposed schedule.

FR6.8

The system shall minimize unnecessary modifications to unaffected tasks.

FR6.9

The system shall update all dependent reminders after rescheduling.

FR6.10

The system shall regenerate the optimized schedule after every accepted modification.